

- Q.27 Explain Enthalpy-Entropy (H-S) diagram.
 Q.28 Describe the low temperature insulation in brief.
 Q.29 Explain dryness fraction and specific volume of steam.
 Q.30 Discuss the role of Economizer and preheater accessories in boiler.
 Q.31 Describe the working of forced draft cooling tower.
 Q.32 Describe the important factors which govern the selection of a refractory.
 Q.33 Explain the safe working properties of refrigerants.
 Q.34 Write the properties and used of ammonia refrigerants.
 Q.35 Differentiate between temporary and permanent hardness of water.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Discuss the construction and working of Cochran boiler in detail with the help of neat diagram.
 Q.37 What are Zeolites? Describe Zeolite process for water softening in detail with the help of neat diagram.
 Q.38 Write short notes on any two of the following
 a) Caustic embrittlement
 b) Boiler corrosion
 c) Properties of Insulation
 d) Selection of refrigerants

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Roll No.

6th Sem /

Chem, P&P, Chem Engg. (Spl. Paint Tech)

Sub : Process Plant Utilities/Proc. Utilities

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Molecular formula of Soda ash is
 a) Na_2CO_3 b) $\text{Al}_2(\text{SO}_4)_3$
 c) $\text{Mg}(\text{OH})_2$ d) $\text{Ca}(\text{OH})_2$
- Q.2 PPM stands for
 a) Permanent Parts Mixture
 b) Parts per million
 c) Part per mixture
 d) Per part mixture
- Q.3 Which of the following is an example of a nature refractory?
 a) Dolomite b) Magnesite
 c) Silica d) Chromite
- Q.4 Which is the following is external treatment process of water softening?
 a) Carbonate conditioning
 b) Colloidal conditioning

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- c) Ion Exchange Process
- d) Calgon conditioning
- Q.5 Which of the following is not a refrigerant?
 - a) Ammonia
 - b) Carbon dioxide
 - c) Cryogeme
 - d) Oxygen
- Q.6 Natural Zeolites are _____
 - a) Porous
 - b) Amorphous
 - c) Non-Durable
 - d) Posses gel structure
- Q.7 Which of the following is a water tube boiler?
 - a) Lancashire boiler
 - b) Cochran boiler
 - c) Babcock & Wilcox boiler
 - d) Locomotive boiler
- Q.8 Hardness of water is due to the presence of salts of
 - a) Boron
 - b) Calcium
 - c) Magnesium
 - d) Both (b) & (c)
- Q.9 Device used to increase the temperature of saturated steam without raising its pressure is called?
 - a) Super heater
 - b) Fusible plug
 - c) Blow off cork
 - d) Stop value
- Q.10 Colloidal impurities include
 - a) Bacteria in water
 - b) Organic waste matter in water
 - c) Oxides of nitrogen
 - d) Sodium salt

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define permanent hardness.
- Q.12 List any two causes of boiler corrosion.
- Q.13 Define sensible heat.
- Q.14 What are primary refrigerants?
- Q.15 Define de mineralized water.
- Q.16 Name a cold insulating material used in the industry.
- Q.17 Write any one unit of hardness.
- Q.18 Define super heated steam.
- Q.19 Define refractoriness.
- Q.20 Write Chemical formula of EDTA.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Discuss in brief the failure of refractories.
- Q.22 Explain in brief the disadvantages of hard water.
- Q.23 Discuss the lime-soda process of water softening in brief.
- Q.24 Define steam trap. Discuss the used and types of steam trap.
- Q.25 Explain in brief difference between priming and foaming.
- Q.26 Differentiate between fire clay and silica refractories.

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